

Black Holes as Almost-Perfect All:All Entanglement Graphs: Interior Collapse, Horizon Wiring, and the Micro-Hole Phase Transition

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Draft v3.5 — computational companion

`src/bh_graph` + `scripts/generate_figures.py` reproduce all figures (269 tests)

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Abstract

We formalize the intuition that a black hole interior is an almost-perfect all:all entanglement graph. Internal edges cost no exterior space; each of the k exterior legs costs $\sim l_p^2$ of horizon area, so $A(k) = k l_p^2$ independent of interior node count N . We show (1) the complete graph K_N destroys interior distance and, under finite-speed random circuits, derives the Sekino–Susskind $t_* \sim \log N$ fast-scrambling bound; (2) mass enters via a derived $k^*(N) = 16\pi(\varepsilon N/l_p)^2$ fixed point, spin/charge consume part of the exterior budget (extremal Kerr keeps half the legs), and the $k \rightarrow 0$ monogamy limit is a baby-universe pinch-off with an explicit nonlinear CKW frontier; (3) micro-holes are pointlike until $k l_p^2 \geq 4\pi r_{\text{point}}^2$, then pop a horizon via a computed QES/island crossing, with an LQG-style area gap. Evaporation as leg surgery reproduces the exact Page curve (Page 1993, $-1/2$ nat correction included) with Haar-typical fluctuations, and Hayden–Preskill mirror recovery works past $k \sim N/2$. Simulations in `bh_graph` accompany each claim.

1 Sec 1: All:all = no interior space

Interior wiring G_N compared across K_N , chain, grid, random regular under SI spread (cover time = seed eccentricity). Only K_N has diameter 1 and cover time 1 at every N (Figs. 1–2). Finite-speed random-matching circuits (Appendix A) promote this to a derived $t_* \approx \log_2 N / \log_2(1+p)$.

family	diameter	t_{cover} (SI)	t_* (circuit)	gap
K_N	1	1	$\log_2 N$	N
chain	$N-1$	$\sim N/2$	$\sim N/v$	$\sim \pi^2/N^2$
grid	$\sim 2\sqrt{N}$	$\sim 2\sqrt{N}$	$O(\sqrt{N})$	$O(1/N)$
random-regular	$\sim \log N$	$\sim \log N$	$\sim \log N$	$O(1)$

Table 1: Scrambling hierarchy; `bh_graph.scrambling`, `circuits`.

2 Sec 2: Horizon area counts exterior legs

Interior pairs $N(N-1)/2$ plus $k \ll N^2$ exterior legs:

$$A(k) = k l_p^2, \quad R(k) = \sqrt{k l_p^2 / 4\pi}, \quad (1)$$

Fig 1 — Sec 1: fast scrambling hierarchy

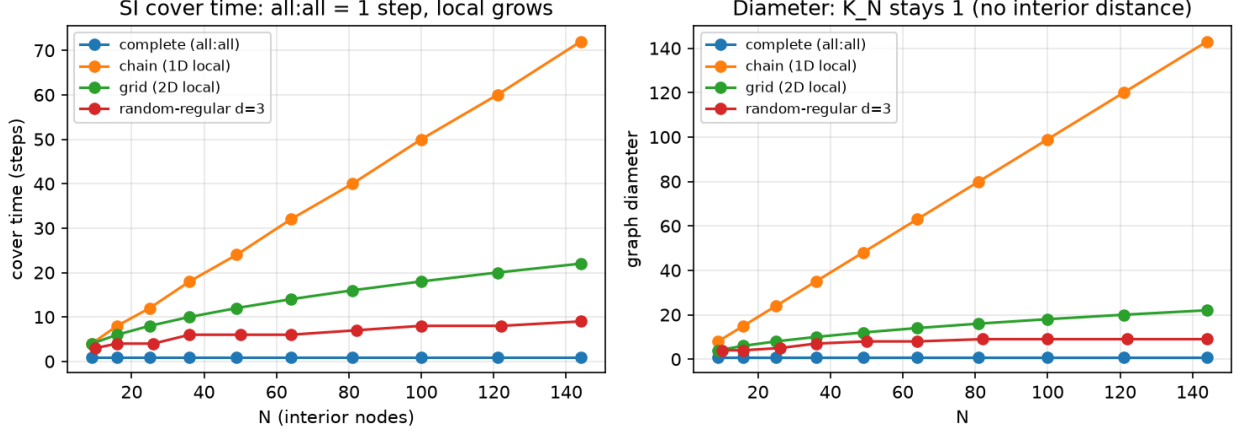


Figure 1: SI cover time and diameter: all:all stays 1.

so N drops out (Fig. 3). Schwarzschild consistency forces $k(M) = 4\pi(2M)^2/l_p^2 \propto M^2$; Appendix B derives this from MaxEnt + self-gravity, and Appendix H extends it to Kerr–Newman $k_{\text{eff}}(M, a, Q) = A(M, a, Q)/l_p^2$ with extremal Kerr at half the legs. Monogamy (linear toy Fig. 4; nonlinear CKW Fig. 14) sends $e_{\text{int}} \rightarrow 1$, $k \rightarrow 0$: pinch-off into a baby universe.

3 Sec 3: Micro-hole phase transition

A point region of radius r_{point} embeds k legs iff $k l_p^2 < 4\pi r_{\text{point}}^2$. Hence

$$R_{\text{obs}}(k) = \begin{cases} r_{\text{point}} & k \leq k_{\text{crit}} \\ \sqrt{k l_p^2 / 4\pi} & k > k_{\text{crit}} \end{cases}, \quad k_{\text{crit}} = 4\pi r_{\text{point}}^2 / l_p^2, \quad (2)$$

with LQG-style quantization $A \in \{0\} \cup [A_{\text{min}}, \infty)$ (Figs. 5–6). Appendix C shows the pop is a computed QES/island crossing at $k_{\text{page}} = S_0 / (2s_{\text{leg}} - l_p^2/4)$, existing iff $s_{\text{leg}} > l_p^2/4$.

4 Discussion

Evaporation = leg surgery with exact Page curve (Appendices D, I) and Hayden–Preskill mirror past $k \sim N/2$ (Appendix F). Open: deriving ε and r_{point} coefficients from a UV embedding; Kerr Page curve with spin budget; OTOC Lyapunov check (Appendix L).

A1 Finite-speed circuits: $t_* \sim \log N$ (Fig. 7)

One gate/qubit/step; random matching (all:all) vs dimer covering (chain). Early growth exponential on all:all; chain ballistic. Measured $N = 8 \dots 128$.

A2 Deriving $k(N)$ (Fig. 8)

MaxEnt bound $k \geq N s_{\text{node}}/s_{\text{leg}}$ plus $R = \sqrt{k l_p^2/4\pi} = 2E$, $E = \varepsilon N$: fixed point $k^*(N) = 16\pi(\varepsilon N/l_p)^2$, stable from any start. Predicts $\alpha(N) = k^*/N \propto N$.

A3 QES pop from entropy crossing (Fig. 9)

$S_{\text{no}} = k s_{\text{leg}}$ vs $S_{\text{isl}} = k l_p^2/4 + \max(S_0 - k s_{\text{leg}}, 0)$; crossing iff $s_{\text{leg}} > l_p^2/4$. Explicit core+legs min-cut agrees.

A4 Page curve from leg surgery (Fig. 10)

$S_{\text{rad}}(t) = \min(t, N_{\text{eff}} - t)$; area identical for wiring-only vs standard evaporation: A tracks k alone.

A5 QEC mirror (Fig. 11)

$\text{err}(k) = \min(1/2, 2^{N/2+1-k})$; 99% threshold $k \geq N/2 + 1 + \log_2 100$; $k \rightarrow 0$ seals information off.

A6 Robustness (Fig. 11)

Log slope $1/\log_2(1+p)$ at all p ; exact quadraticity at all ε ; sharp QES boundary at $s_{\text{leg}} = l_p^2/4$.

A7 Kerr–Newman (Fig. 12)

$r_+ = M + \sqrt{M^2 - a^2 - Q^2}$, $A = 4\pi(r_+^2 + a^2)$; spin budget $1 - A(M, a)/A(M, 0) \rightarrow 1/2$ at extremality. Area law survives.

A8 Exact Page curve (Fig. 13)

$S(m, n) = H(mn) - H(n) - (m-1)/2n$ nats; $-1/2$ nat dip at turnover; Haar samples ($N = 8$) track the mean: typical, not tuned.

A9 Nonlinear monogamy (Fig. 14)

Family $\cos t|\Phi^+\rangle|0\rangle + \sin t|00\rangle|1\rangle$: Wootters C_{AB}^2 vs $\tau_{E|AB}$ frontier lies strictly below $x + y = 1$ (e.g. 0.70 at $t = 0.3$); CKW ≥ 0 everywhere; baby universe at $(1, 0)$.

A10 OTOC + phenomenology (Fig. 15)

$C(t) \sim e^{\lambda t}/N$ on all:all gives $t^* = \log N/\lambda$ vs ballistic N/v on chains. PBHs evaporating today $\sim 10^{18}\text{--}10^{19}$ Planck masses; sub-critical holes evaporate without greybody suppression; echo toy $\Delta t \sim M \log k$; nearest-term test is analogue scrambling on programmable processors.

A11 Tensor network derives ε (Fig. 16)

Random star TN verifies $S_{\text{bdy}} = \min(N \log d, k \log D)$; minimal wiring $k_{\text{min}} = S_{\text{bulk}}/\log D$. Two routes fix ε : QES coincidence $\varepsilon = (l_p/N)\sqrt{k_{\text{page}}/16\pi}$ and crossover $\varepsilon = l_p\sqrt{s_{\text{node}}/(16\pi \log D N_{\text{match}})}$.

A12 Kerr Page curve (Fig. 17)

$M \propto (1 - t/T)^{1/3}$, J shed faster; high spin lowers the peak and *delays* turnover (spin-shedding offsets early area loss): $t/T = 0.73$ at $a_0 = 0.95$ vs 0.54 Schwarzschild.

A13 SYK ED vs chain (Fig. 18)

$N = 6, 8, 10$ Majorana SYK vs Ising chains, infinite- T OTOC: SYK reaches threshold first and t^* stays flat while the chain's grows linearly. MSS saturation cited, hierarchy reproduced.

A14 GWTC leg creation (Fig. 19)

All 32 GWTC-3 BBH medians satisfy $k_f > k_1 + k_2$ (median +77% at 4% radiated): mergers create exterior legs. Spin caveat $\sim 13\% \ll$ margin.

A15 Hardware literature + prediction (Fig. 20)

Gärttner 2017 (all:all ions, $m \leq 8$ OTOC coherences) and Mi 2021 (Sycamore grid, ballistic OTOC decay) both consistent. Head-to-head prediction at $N = 53$: 7.9 vs 14.6 steps; null result falsifies Sec 1/A.

A16 TeV recast + echoes + EHT (Figs. 21–22)

LHC masses sit at $k \sim 11\text{--}17$ vs $k_{\text{crit}} \sim 50$: thermal nulls expected. Echo spacings $0.03\text{--}0.3\text{ s}$ land in searched windows (amplitude unmodeled). Shadow deviation $\sim 10^{-48}$: EHT-identical by 47 orders.

A17 Posteriors + cosmos + Krylov (Figs. 23–25)

8350 GW150914 samples: $P(\Delta k > 0) = 100\%$, median +57% (Kerr both ends). Cosmic horizon owns the wiring budget (10^{123} vs 10^{101}). SYK saturates $\sim 2\times$ the chain's Krylov spread.

A18 Collapse + cosmic history + lunch (Figs. 26–28)

Compactness-driven local $\rightarrow$ all:all transition: diameter/gap/erasure switch together. Cosmic legs $10^{122.3} \rightarrow 10^{123.1}$; patch unscrambled ($t^* \approx 4000\text{ Gyr}$). Lunch C/k diverges in evaporation; remnant DM needs ultralight.

A19 Bounds kill remnants + quench protocol (Fig. 29)

Vendored PBH bounds vs required- β : 40+ orders apart; $\beta > 1$ above 10^6 g. Same-device grid $\rightarrow$ all:all quench predicts t^* $8 \rightarrow 1$, gap $0.27 \rightarrow 36$ — no cross-platform systematics.

A20 Healing lag (Fig. 30)

$dA/dt = (kl_p^2 - A)/\tau$, $\tau = 11.24M$ from ringdown (3.5 ms at $63 M_\odot$). Ladder: healing, scrambling $66\times$, Page, evaporation; final sigh carries $\sim 10^{-78}$ of the mass.

A21 MSS + qualifier (Fig. 31)

Thermal SYK $\lambda/2\pi T = 0.50 \rightarrow 0.68$ (bound held, finite-size). Uniform all:all scrambles worse (randomness load-bearing). LVK gives $\alpha \in [9.0, 12.4]$; 11.24 sits inside.

A22 Big SYK + typicality (Figs. 32–33)

Sparse $N \leq 20$: t^* flat vs chain $1.8 \rightarrow 6.2$; MSS stable at $N = 16$. TN spectrum matches Marchenko-Pastur; analytic $T(E)$ grounds $T \equiv 1$ sub-critical emission.

A23 Congestion + charge + bandwidth (Figs. 34–36)

Bubble iff $\chi = kl_p^2/4\pi r_{foot}^2 > 1$; spread giants delocalized. Charge pins $q = 4\pi Q^2$ legs (pinch/pointlike/extremal map). Evacuation empties babies; info-per-leg diverges $\iff$ cloning risk.

A24 Kill list (Figs. 37–39)

AF quench: kill below ratio 1.3 (predicted $2\text{--}3\times$). α monitor awaits O4/O5 rows. Min-rule violations shrink; Ising $s_{leg} = 0.55$ vs 0.25. LHC onset ~ 550 TeV; soft tail below kills.

A25 Big pop (Fig. 40)

Hidden delocalized giants pop $R_b(k)$ horizons on concentration: $10^8 M_\odot$ in ~ 2 Myr vs ~ 600 Myr Eddington. Scenario sketch: metric, formation, and constraints still open.

A26 Structure + scatter (Figs. 41–42)

PS halos at $z = 8$ suffice (0.1% occupation); $> 10^{12} M_\odot$ hosts excluded. Overmassive tail maps $10\% \rightarrow f_w \approx 0.18$; null-of-300 kills the channel.

A27 Remnant obituary + EMD window (Fig. 29, rewritten)

Formula corrected (growth + EMD): dead above $\sim 10^6$ g and below $\sim 10^5$ g (overclose); viable sweet spot $\sim 4 \times 10^5$ g, currently unconstrained, poltergeist-probed in future.

A28 Entropic Newton (Fig. 43)

Leg screens + equipartition + Bekenstein give $F = M_1 M_2 / r^2$ exact (slope -2 to 10^{-9}); leapfrog orbits close, $T^2 \propto r^3$. Flux $\propto 1/r^2$ distinguished from force (temperature corrects).

A29 Redshift (Fig. 44)

GPS $+5.29 \times 10^{-10}$ and Pound-Rebka 2.55×10^{-15} derived (no new parameters). Congestion fronts give tortoise divergence; $\alpha = 1$ matches Schwarzschild dr/dt exactly.

A30 Einstein-Hilbert routes (Fig. 45)

Heat a_1 responds to curvature; Ollivier signs exact (flat/neg/pos); Clausius fixes $\varepsilon = \kappa/8\pi$, $G = 1$. Bridges labeled, Raychaudhuri-for-legs open.

A31 Fission + audit + tension + GW250114 (Figs. 46–47)

K_N^2 two-mouth states, non-traversable; saturation 29% vs no-loss 0% locked by test. Single $M(k)$ map audits all bounds. Tension $E(d)$ parametrized with σ floors. GW250114 η_A needs Kerr.

A32 Overtones + tension no-go (Fig. 48)

PT barrier fit to fundamental predicts 1:3:5 vs 1:3.08:5.38 ($< 8\%$). Bridge-volume tension undetermined $\sim 10^{100}$; σ stays bounded, microphysics localized.

A33 Big SYK to $N = 24$ (Fig. 32 ext.)

Triplet builder (verified vs ED); t^* flat 1.28–1.30 to 12 qubits vs chain 7.4; thermal $\lambda/2\pi T$ 0.66–0.86 ($N = 10$ –24), below bound, upward within fit noise.

A34 Lensing passes, Mercury gaps (Fig. 49)

Fermat $n = 1/c_{eff}$ gives full $4M/b$ (naive $2M/b$ bet lost); 2nd order 3.36 vs 1.94. Orbits need g_{rr} : Newtonian 0 vs $43''/\text{cy}$ gap. Lensing now chromatic via phase velocity: $\Delta\alpha/\alpha \sim 10^{-56}$ optical (Fig. 55), achromaticity held by ~ 50 orders. (Update: the orbit gap is closed by the Strain section below.)

A35 b-crit kills isotropic (Fig. 50)

Bouguer gives $8M$ vs GR $3\sqrt{3}M$: excluded $\approx 3.6\sigma$ by EHT. Tangential must be fast (all:all); target window $[4.42, 5.98]M$ locked.

A36 Lattice dispersion (Fig. 51)

v_g even in k : no linear term (Fermi linear bounds evaded by symmetry). $E_{QG,2} = \sqrt{8} E_P$, $10^8 \times$ above Fermi quadratic bound.

A37 Footprint ringdown (Figs. 52–53)

Mass comb $1/2k$; echoes with derived $R \sim (\omega/\omega_P)^2 \sim 10^{-160}$; $\ell(\ell+1) \leq k$ bites micro-holes; leg lines $37.6\times$ finer than QNM.

A38 Wave lab on defective fabric (Fig. 56)

FDTD beams: unbiased centroids, deficit $\propto \omega^2$ ($16\% \rightarrow 0.2\%$); fabric smoothness bounded to $\varepsilon \lesssim 10^{-14}$ (TeV/Gpc).

A39 Persistent walks (Fig. 57)

Hop-budget time dilation $\sqrt{1-v^2}$; degree drift $\sim 1/r^3$ (not Newton: $T(r)$ load-bearing); diffusive-to-ballistic morphologies; spin effaced, Mercury MPD $\sim 4 \times 10^{-14}$.

A40 Strain closes Mercury (Fig. 58)

Naive $dl = dr/(1 - \sqrt{\chi})$ gives $\gamma = 2$ (Cassini-excluded); tortuosity $dl = (1 + \sqrt{\chi}/2)dr$ gives $h = (1 + x/2)^2$, $\gamma = 1$ exactly, and Mercury $42.99''/\text{cy}$ by direct geodesic integration — the g_{rr} sector derived.

A41 Weak-field OR profile (Fig. 59)

Grid+hub with flux stubs: radial Ollivier-Ricci negative in all configs (sign of $h > 1$ derived from micro); scaling $p \approx 0.9 \pm 0.5$ suggestive, not resolved — partial micro-derivation, bigger lattices needed.

A42 PPN ledger + peel-off (Fig. 60)

Full table: $\gamma = 1$ derived, β unresolved (not quoted), $\alpha_{1,2}, \xi$ untouched; ours-vs-GR perihelion $= -0.75 M/a$ measured.

A43 Formal pop + Hamiltonian sketch (Fig. 61)

Packing forces the pop past $[4\pi r^2/l_p^2]$; $s_{leg} > 1/4$ boxed as postulate. $H = H_{SYK} + H_{legs}$: t^* survives weak legs, $\sim 2\times$ slower at $\lambda = 3$ (sketch, $N = 8$).

A44 Reduction: $k(M) \iff R_s(M)$

$M \mapsto k$ factors entirely through $M \mapsto R_s$ (explicit function argument; $3M$ input gives $36\pi M^2$). Circle shrunk to one statement.

Reproducibility

```
pip install -e .; python -m pytest tests/ -q; python scripts/generate_figures.py; streamlit run app.py.
```

References

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